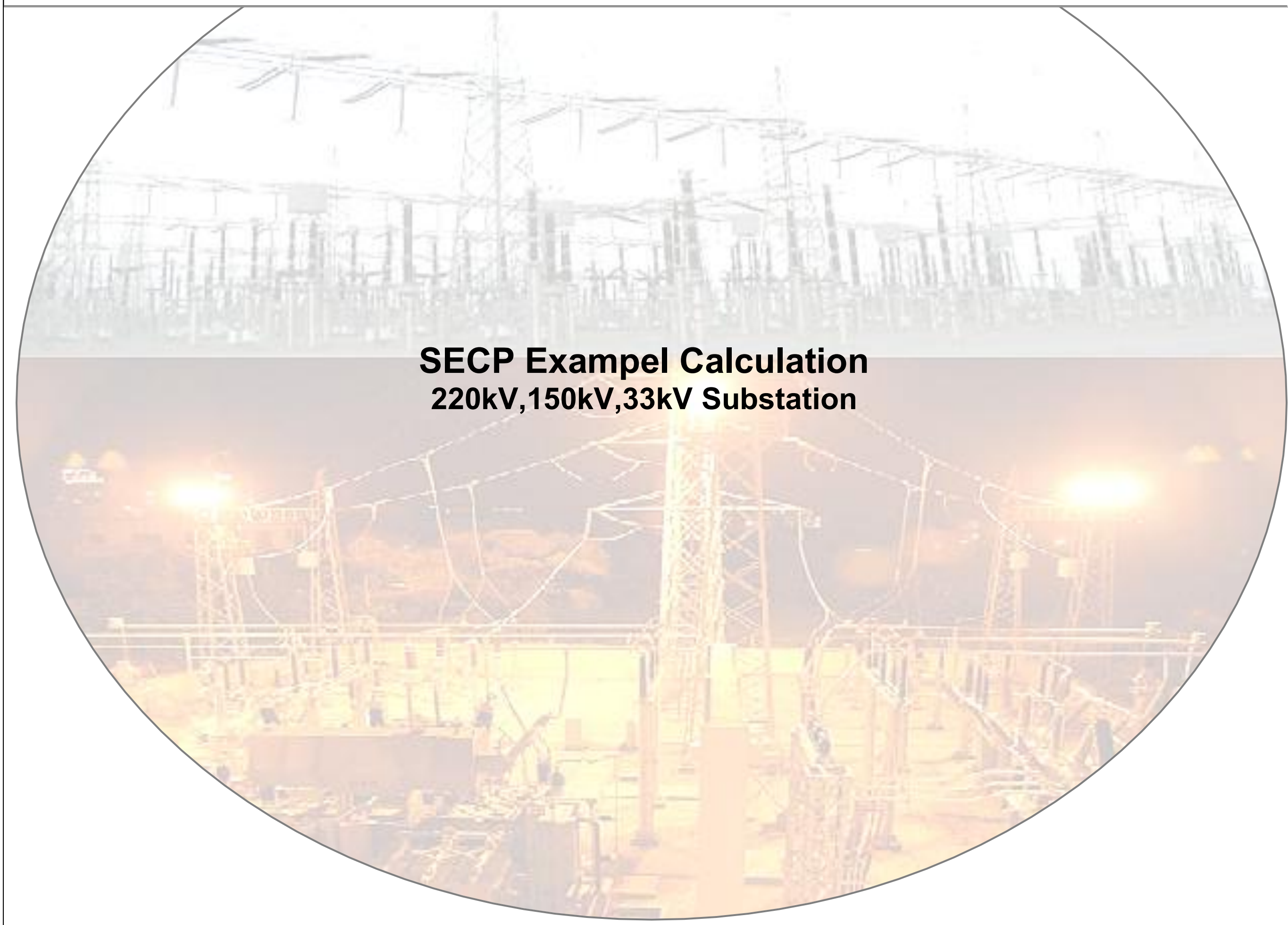
CT AND VT ADEQUACY CHECK CALCULATION



SECP Exampel Calculation
220kV,150kV,33kV Substation

GLOBAL SECP PROGRAM
Output Report– 06.08.2026

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Output report - CT and VT adequacy check calculation					
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Project Id / Number	:	112358132134
Project Name	:	SECP Exampel Calculation
Substation Name	:	Example
Customer Name	:	Customer1
Country	:	Netherlands
Voltage Level	:	220kV
List of bays	:	Line Diff - 220
		Trafo Diff - 220
		Over Current - 220
Voltage Level	:	150kV
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Voltage Level	:	33kV
List of bays	:	Line Diff - 33
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Document Information:

Revision	:	Revision - 2
Document Number	:	
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Reference standard:

Current Transformer:

	Description	Standard reference
.	General requirements for Instrument Transformers	IEC 61869-1, Edition 1.0,2007-10
.	Instrument transformers - Part 2: Additional requirements for current transformers	IEC 61869-2, Edition 1.0,2012-09
.	Instrument transformers - Part 100: Guidance for application of current transformers in power system protection	IEC 61869-100, Edition 1.0, 2017-01

Nomenclature:

CT Nomenclature:

	Description	SI Unit	Symbol
	Nominal System voltage	kV	U_n
	System frequency	Hz	f_r
	Primary time constant of source impedance	ms	T_p
	Rated primary current of the transformer	A	I_{nt}
	Maximum primary fundamental frequency current that passes two main CTs and the transformer	A	I_{tf}
	Through fault MVA of transformer	MVA	T_{MVAe}
	Internal burden of the current transformer	VA	S_{in}
	Actual burden of the current transformer	VA	S_a
	Maximum close-in fault current	A	$I_{k\,max}$
	Maximum primary fundamental frequency 1-phase fault current for external through fault current	A	$I_{te\,max}$
	Maximum primary fundamental frequency 3-phase fault current for external through fault current	A	$I_{t\,max}$
	Maximum primary fundamental frequency 1-phase fault current for internal close-in fault current	A	$I_{ke\,max}$
	Maximum primary fundamental frequency 3-phase fault current for internal close-in fault current	A	$I_{k\,max}$
	Maximum primary fundamental frequency current that passed two main CTs without passing the transformer	A	I_f
	Maximum primary fundamental frequency through fault current that passes two main CTs (one-and-a-half or double-breaker) without passing the protected line	A	I_{tfdb}
	Primary operate value	A	I_{op}
	Transformer impedance at principal tap		Z_t
	Transformer rating	MVA	T_{MVA}
.	Transformer impedance at principal tap	%	Z_t

Scenario 220kV

General data					
	Type of switchyard (AIS/GIS)		=	AIS	
	Nominal System voltage	U_n	=	220	kV
	System frequency	f_r	=	50	Hz
	Maximum Conductor temperature	°c	=	75	deg

Bay ID: Line Diff - 220(LD - 220)

Core ID: Core-1

Application Name: LineDifferentialProtection

Relay Input data					
	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency 3-phase fault current for internal close-in fault current	$I_{k\,max}$	=	50000	A
	Maximum primary fundamental frequency 1-phase fault current for internal close-in fault current	$I_{ke\,max}$	=	50000	A
	Maximum primary fundamental frequency 3-phase fault current for external through fault current	$I_{t\,max}$	=	50000	A
	Maximum primary fundamental frequency 1-phase fault current for external through fault current	$I_{te\,max}$	=	50000	A
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Power Transformer included in the protection zone	YES/NO	=	No	

Core data					
	Core Details		=	Core-1	
	Core Application		=	LineDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RED670	
	Relay Burden	S_R	=	0,02	VA

	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	90	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	800	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	7	Ohm
·	Rated output of the Current Transformer	S_n	=	30	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$**:**

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0C * (1 + (Max.\, conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max.\, conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((5{,}6065) * 90/1\right)}{1000}$$

$$R_{\text{lead}} = 1{,}0092\ \ Ohm$$

Relay Parameters Calculation

$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$

$I_{\mathrm{R}} = 1\ A$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\mathrm{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\mathrm{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\mathrm{relay}} = 0{,}02\ \ Ohm$$

Tap ID : 1 (LineDifferentialProtection)

CT knee point voltage Calculation as per Manual
(Ref: Annexure for relay manual)

$$Equivalent\ secondary\ e.m.f.for\ 3 - ø\ internal\ close - in\ faults\ (E_{3-p\ in}) = I_{k\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + R_{lead} + R_{relay})$$

$$E_{3-p\ in} = 50000 * \left(\frac{1}{800}\right) * (7 + 1{,}0092 + 0{,}02)$$

$$E_{3-p\ in} = 501{,}825\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1 - ø\ internal\ close - in\ faults\ (E_{1-p\ in}) = I_{ke\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + 2R_{lead} + R_{relay})$$

$$E_{1-p\ in} = 50000 * \left(\frac{1}{800}\right) * (7 + (2 * 1{,}0092) + 0{,}02)$$

$$E_{1-p\ in} = 501{,}825\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 3 - ø\ external\ through\ faults\ (E_{3-p\ ex}) = 2 * I_{t\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + R_{lead} + R_{relay})$$

$$E_{3-p\ ex} = 2 * 50000 * \left(\frac{1}{800}\right) * (7 + 1{,}0092 + 0{,}02)$$

$$E_{3-p\ ex} = 1003{,}65\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1 - ø\ external\ through\ faults\ (E_{1-p\ ex}) = 2 * I_{te\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + 2R_{lead} + R_{relay})$$

$$E_{1-p\ ex} = 2 * 50000 * \left(\frac{1}{800}\right) * (7 + (2 * 1{,}0092) + 0{,}02)$$

$$E_{1-p\ ex} = 1003{,}65\ V$$

$$Minimum\ required\ Knee\ point\ voltage\ (V_k) = Max(E_{3-p\ in}, E_{1-p\ in}, E_{3-p\ ex}, E_{1-p\ ex})$$

$$V_k = Max\ (501{,}825, \quad 501{,}825, \quad 1003{,}65, 1003{,}65)$$

$$V_k = 1003{,}65\ \ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$CT\ secondary\ limiting\ e.m.f.\ E_{ALF} = ALF * I_{sr} * \left(R_{ct} + \frac{S_n}{I_{sr}^{\ 2}}\right)$$

$$E_{ALF} = 20 * 1 * \left(7 + \frac{30}{1^2}\right)$$

$$E_{ALF} = 740\ V$$

$$E_{rq} = E_{ALF}$$

$$E_{rq} = 740\ V$$

$Proposed\ knee\ point\ voltage\ \left(E_{rq}\right) < Required\ knee\ point\ voltage(V_k)$

$CT\ Tap\ is\ not\ adequate$

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Summary of Application : LineDifferentialProtection

CT-Protection Core

Summary							
Tap Details	***CT Primary current*** $\boldsymbol{I_{pr}\ (A)}$	***CT Secondary*** **current** $\boldsymbol{I_{sr}\ (A)}$	***Accuracy Class***	***Accuracy Limiting Factor*** (ALF)	***Knee point voltage*** (V)		***Results***
					Proposed	***Calculated***	
Tap-1	800	1	5PR	20	740	1003,65	Not Adequate

Bay ID: Trafo Diff - 220(TD - 220)

Core ID: Core-1

Application Name: TransformerDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency phase to earth current that passed two main CTs without passing the	I_{tf}	=	2000	A
	Transformer rating	T_{MVA}	=	220	MVA
	Transformer impedance at principal tap	Z_t	=	23	%
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Type of System	$Solidly$ $/High\ impedance$	=	Solidly	

Core data

	Core Details		=	Core-1	
	Core Application		=	TransformerDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RET670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	90	m
	Cable Conductor Type		=	Copper	

	Current transformer Input data				
	Description	**Symbol**		**Value**	**SI Unit**
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	800	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	7	Ohm
·	Rated output of the Current Transformer	S_n	=	30	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$**:**

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0 C * (1 + (Max.\,conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max.\,conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((5{,}6065) * 90/1\right)}{1000}$$

$$R_{\text{lead}} = 1{,}0092\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$$

$$I_{\text{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\text{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\text{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\text{relay}} = 0{,}02\ \ Ohm$$

Tap ID : 1 (TransformerDifferentialProtection)

Model:RET670, RET650

(Ref: ABB application manual, RET-670, section-2 requirements, Cl.no:2.1.6.1)

Fault Calculations

$$Rated\ primary\ current\ of\ the\ transformer\ (\mathrm{I_{nt}})\ =\ \frac{\mathrm{T_{MVA}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$$

$$\mathrm{I_{nt}}\ =\ \frac{220 * 1000000}{220 * \sqrt{3} * 1000}$$

$$\mathrm{I_{nt}} = 577{,}3503\ \mathrm{A}$$

$$Through\ fault\ MVA\ of\ the\ Transformer\ \mathrm{T_{MVAe}} = \frac{\mathrm{T_{MVA}} * 100}{\mathrm{Z_t}}$$

$$\mathrm{T_{MVAe}} = \frac{220 * 100}{23}$$

$$\mathrm{T_{MVAe}} = 956{,}5217\ MVA$$

$$Maximum\ primary\ fundamental\ frequency\ current\ that\ passes\ two\ main\ CTs\ and\ the\ transformer\ (\mathrm{I_{tf}}) = \frac{\mathrm{T_{MVAe}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = \frac{956{,}5217 * 1000000}{220 * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = 2510{,}2185\ A$$

CT knee point voltage Calculation as per Manual

$$Equivalent\ secondary\ e.m.f.for\ close-in\ faults\ (E_{alreq1}) = 30 * I_{nt} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$$

$$E_{alreq1} = 30 * 577{,}3503 * \frac{1}{800} * (7 + (2 * 1{,}0092) + 0{,}02)$$

$$E_{alreq1}\ = 195{,}6871\ V$$

$$Equivalent\ secondary\ e.m.f.for\ External\ faults\ (E_{\mathrm{alreq2}}) = 2 * I_{tf} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$$

$$E_{\mathrm{alreq2}}\ = 2 * 2510{,}2185 * \frac{1}{800} * (7 + (2 *\ 1{,}0092) + 0{,}02)$$

$$E_{\mathrm{alreq2}} = 56{,}7209\ V$$

$$Maximum\ secondary\ limiting\ e.m.f\ (V_k) = Max(E_{alreq1}, E_{alreq2})$$

$$V_k = Max(195{,}6871{,}56{,}7209)$$

$$V_k = 195{,}6871\ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$Secondary\ limiting\ e.m.f.\ \ E_{ALF}\ = ALF\ \ * I_{sr}\ * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$$

$$E_{ALF} = 20 * 1 * \left(7 + \frac{30}{1^2}\right)$$

$$E_{ALF} = 740\,V$$

$$E_{rq} = E_{ALF}$$

$$E_{rq} = 740\,V$$

$Proposed\ knee\ point\ voltage\ (E_{rq}) > Required\ knee\ point\ voltage (V_k)$

$CT\ Tap\ is\ adequate$

.

Summary of Application : TransformerDifferentialProtection

CT-Protection Core

Summary							
Tap Details	***CT Primary current*** $\boldsymbol{I_{pr}\ (A)}$	***CT Secondary current*** $\boldsymbol{I_{sr}\ (A)}$	***Accuracy Class***	***Accuracy Limiting Factor*** $(\boldsymbol{ALF})$	***Knee point voltage***$(\boldsymbol{V})$		***Results***
					Proposed	***Calculated***	
Tap-1	800	1	5PR	20	740	195,6871	Adequate

Bay ID: Over Current - 220(OC - 220)

Core ID: Core-1

Application Name: OverCurrentProtection

Relay Input data					
	Maximum close-in fault current	$I_{k\,max}$	=	31500	A
	Primary operate value	I_{op}	=	1000	A

Core data					
	Core Details		=	Core-1	
	Core Application		=	OverCurrentProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	REF630	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	90	m
	Cable Conductor Type		=	Copper	

Current transformer Input data					
	Description	**Symbol**		**Value**	**SI Unit**
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	2500	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	14	Ohm
·	Rated output of the Current Transformer	S_n	=	30	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$:

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0C * (1 + (Max.conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max.conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((5{,}6065) * 90/1\right)}{1000}$$

$$R_{\text{lead}} = 1{,}0092\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$$

$$I_{\text{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\text{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\text{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\text{relay}} = 0{,}02\ \ Ohm$$

Tap ID : 1 (OverCurrentProtection)

Preliminary Calculations

$$Internal\ burden\ of\ the\ current\ transformer\ (S_{in}) = \ I_{sr}{}^2 * R_{ct}$$

$$S_{in} = 1^2 * \ 14\ \ VA$$

$$S_{in} = 14\ \ VA$$

$$Actual\ burden\ of\ the\ current\ transformer\ \ (S_a) = \ (I_{sr}{}^2 * R_{lead}) + S_R$$

$$S_a = (1^2 * \ 1{,}0092\ ) + \ 0{,}02$$

$$S_a = 1{,}0292\ \ VA$$

$$Rated\ accuracy\ limit\ factor\ at\ the\ rated\ burden\ of\ the\ CT\ \ (F_n) = \text{ALF}$$

$F_n = 20$

Calculation as per Manual

$To\ ensure\ correct\ operation\ the\ relay\ below\ condition\ need\ to\ satisfy$

$Approximate\ value\ of\ the\ accuracy\ limit\ factor\ F_a > 20 * \frac{I_{op}}{I_{pr}}\ shall\ be\ satisfied$

$Approximate\ value\ of\ the\ accuracy\ limit\ factor\ corresponding\ to\ the\ actual\ CT\ burden\ (F_a) = F_n * (\frac{S_{in} + S_n}{S_{in} + S_a})$

$F_a = 20 * \frac{14 + 30}{14 + 1{,}0292}$

$F_a = 58{,}5527$

$S_n = 30\ VA$

$S_a = 1{,}0292\ VA$

$F_a = 58{,}5527$

$20 * \frac{I_{op}}{I_{pr}} = 8$

$Conditions\ S_n > S_a\ and\ F_a > 20 * \frac{I_{op}}{I_{pr}}\ Satisfied\ ,Hence\ CT\ tap\ is\ adequate$

.

Summary of Application : OverCurrentProtection

CT-Protection Core

Summary							
Tap Details	***CT Primary current*** I_{pr} (A)	***CT Secondary current*** I_{sr} (A)	***Accuracy Class***	***Accuracy Limiting Factor*** (ALF)	***Burden***(VA)		***Results***
					Proposed	***Calculated***	
Tap-1	2500	1	5PR	20	30	1,0292	Adequate

Scenario 150kV

General data				
Type of switchyard (AIS/GIS)		=	AIS	
Nominal System voltage	U_n	=	150	kV
System frequency	f_r	=	50	Hz
Maximum Conductor temperature	°c	=	75	deg

Bay ID: Line Diff - 150(LD - 150)

Core ID: Core-1

Application Name: LineDifferentialProtection

Relay Input data				
Primary time constant of the source (three-ph)	T_p	=	110	ms
Maximum primary fundamental frequency 3-phase fault current for internal close-in fault current	$I_{k\,max}$	=	6600	A
Maximum primary fundamental frequency 1-phase fault current for internal close-in fault current	$I_{ke\,max}$	=	6600	A
Maximum primary fundamental frequency 3-phase fault current for external through fault current	$I_{t\,max}$	=	50000	A
Maximum primary fundamental frequency 1-phase fault current for external through fault current	$I_{te\,max}$	=	50000	A
Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
Power Transformer included in the protection zone	YES/NO	=	No	

Core data				
Core Details		=	Core-1	
Core Application		=	LineDifferentialProtection	
Accuracy Class		=	5PR	
No. of taps on CT Primary	n	=	1	nos

	Relay Make		=	ABB	
	Relay Model		=	RED670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	65	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	1250	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	7	Ohm
·	Rated output of the Current Transformer	S_n	=	10	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$:

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0C * (1 + (Max.\ conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{((Lead\ resistance\ at\ Max.conductor\ temperature.) * Main\ cable\ length/n_1)}{1000})$$

$$R_{\text{lead}} = \ 2 * \frac{((5{,}6065) * 65/1)}{1000}$$

$$R_{\text{lead}}\ = 0{,}7288\ \ Ohm$$

Relay Parameters Calculation

$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\mathrm{R}}) = I_{\mathrm{sr}}$

$I_{\mathrm{R}} = 1\ A$

$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\mathrm{relay}}) = \frac{S_R}{I_R^2}$

$R_{\mathrm{relay}} = \frac{0{,}02}{1^2}$

$R_{\mathrm{relay}}\ = 0{,}02\ \ Ohm$

Tap ID : 1 (LineDifferentialProtection)

CT knee point voltage Calculation as per Manual

(Ref: Annexure for relay manual)

$Equivalent\ secondary\ e.m.f.for\ 3 - ø\ internal\ close - in\ faults\ (E_{3-p\ in}) =\ I_{k\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} +\ R_{lead} + R_{relay})$

$E_{3-p\ in} = 6600 * \left(\frac{1}{1250}\right) *\ (7 +\ 0{,}7288 +\ 0{,}02)$

$E_{3-p\ in} =\ 40{,}9137\ V$

$Equivalent\ secondary\ e.m.f.for\ 1 - ø\ internal\ close - in\ faults\ (E_{1-p\ in}) =\ I_{ke\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} +\ 2R_{lead} + R_{relay})$

$E_{1-p\ in} = 6600 * \left(\frac{1}{1250}\right) *\ (7 +\ (2 *\ 0{,}7288) +\ 0{,}02)$

$E_{1-p\ in} =\ 40{,}9137\ V$

$Equivalent\ secondary\ e.m.f.for\ 3 - ø\ external\ through\ faults\ (E_{3-p\ ex}) =\ 2 * I_{t\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} +\ R_{lead} + R_{relay})$

$E_{3-p\ ex} = 2 *\ 50000 * \left(\frac{1}{1250}\right) *\ (7 +\ 0{,}7288 +\ 0{,}02)$

$E_{3-p\ ex} =\ 619{,}904\ V$

$Equivalent\ secondary\ e.m.f.for\ 1 - ø\ external\ through\ faults\ (E_{1-p\ ex}) =\ 2 * I_{te\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} +\ 2R_{lead} + R_{relay})$

$E_{1-p\ ex} = 2 *\ 50000 * \left(\frac{1}{1250}\right) *\ (7 +\ (2 *\ 0{,}7288) +\ 0{,}02)$

$E_{1-p\ ex} =\ 619{,}904\ V$

$Minimum\ required\ Knee\ point\ voltage\ (V_k) =\ Max(E_{3-p\ in}, E_{1-p\ in}, E_{3-p\ ex}, E_{1-p\ ex})$

$V_k = Max\ (40{,}9137, \quad 40{,}9137, \quad 619{,}904{,}619{,}904)$

$V_k = 619{,}904\ V$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$CT\ secondary\ limiting\ e.m.f.\ E_{ALF} = ALF * I_{sr} * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$

$E_{ALF} = 20 * 1 * \left(7 + \frac{10}{1^2}\right)$

$E_{ALF} = 340\ V$

$E_{rq} = E_{ALF}$

$E_{rq} = 340\ V$

$Proposed\ knee\ point\ voltage\ \left(E_{rq}\right) < Required\ knee\ point\ voltage(V_k)$

$CT\ Tap\ is\ not\ adequate$

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Summary of Application : LineDifferentialProtection

CT-Protection Core

Summary							
Tap Details	***CT Primary current*** $I_{pr}\ (A)$	***CT Secondary current*** $I_{sr}\ (A)$	***Accuracy Class***	***Accuracy Limiting Factor*** (ALF)	***Knee point voltage***(V)		***Results***
					Proposed	***Calculated***	
Tap-1	1250	1	5PR	20	340	619,904	Not Adequate

Core ID: Core-2

Application Name: LineDifferentialProtection

Relay Input data					
	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency 3-phase fault current for internal close-in fault current	$I_{k\,max}$	=	29000	A
	Maximum primary fundamental frequency 1-phase fault current for internal close-in fault current	$I_{ke\,max}$	=	29000	A
	Maximum primary fundamental frequency 3-phase fault current for external through fault current	$I_{t\,max}$	=	40000	A
	Maximum primary fundamental frequency 1-phase fault current for external through fault current	$I_{te\,max}$	=	40000	A
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Power Transformer included in the protection zone	YES/NO	=	No	

Core data					
	Core Details		=	Core-2	
	Core Application		=	LineDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RED670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	65	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	1250	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	7	Ohm
·	Rated output of the Current Transformer	S_n	=	10	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation ($\mathbf{R}_{lead}$):

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0C * (1 + (Max.conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max.conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\mathrm{lead}} = 2 * \frac{\left((5{,}6065) * 65/1\right)}{1000}$$

$$R_{\mathrm{lead}}\ = 0{,}7288\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\mathrm{R}}) = I_{\mathrm{sr}}$$

$$I_{\mathrm{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\mathrm{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\mathrm{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\mathrm{relay}}\ = 0{,}02\ \ Ohm$$

Tap ID : 1 (LineDifferentialProtection)

CT knee point voltage Calculation as per Manual
(Ref: Annexure for relay manual)

$$Equivalent\ secondary\ e.m.f.for\ 3-ø\ internal\ close-in\ faults\ (E_{3-p\ in}) = \ I_{k\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + \ R_{lead} + R_{relay})$$

$$E_{3-p\ in} = 29000 * \left(\frac{1}{1250}\right) * \ (7 + \ 0{,}7288 + \ 0{,}02)$$

$$E_{3-p\ in} = \ 179{,}7722\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1-ø\ internal\ close-in\ faults\ (E_{1-p\ in}) = \ I_{ke\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + \ 2R_{lead} + R_{relay})$$

$$E_{1-p\ in} = 29000 * \left(\frac{1}{1250}\right) * \ (7 + \ (2 * \ 0{,}7288) + \ 0{,}02)$$

$$E_{1-p\ in} = \ 179{,}7722\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 3-ø\ external\ through\ faults\ (E_{3-p\ ex}) = \ 2 * I_{t\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + \ R_{lead} + R_{relay})$$

$$E_{3-p\ ex} = 2 * \ 40000 * \left(\frac{1}{1250}\right) * \ (7 + \ 0{,}7288 + \ 0{,}02)$$

$$E_{3-p\ ex} = \ 495{,}9232\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1-ø\ external\ through\ faults\ (E_{1-p\ ex}) = \ 2 * I_{te\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + \ 2R_{lead} + R_{relay})$$

$$E_{1-p\ ex} = 2 * \ 40000 * \left(\frac{1}{1250}\right) * \ (7 + \ (2 * \ 0{,}7288) + \ 0{,}02)$$

$$E_{1-p\ ex} = \ 495{,}9232\ V$$

$$Minimum\ required\ Knee\ point\ voltage\ (V_k) = \ Max(E_{3-p\ in}, E_{1-p\ in}, E_{3-p\ ex}, E_{1-p\ ex})$$

$$V_k = Max\ (179{,}7722, \quad 179{,}7722, \quad 495{,}9232, 495{,}9232)$$

$$V_k = 495{,}9232\ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$CT\ secondary\ limiting\ e.m.f.\ E_{ALF} = ALF * I_{sr} * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$$

$$E_{ALF} = 20 * 1 * \left(7 + \frac{10}{1^2}\right)$$

$$E_{ALF} = 340\ V$$

$E_{rq} = E_{ALF}$

$E_{rq} = 340\,V$

$Proposed\ knee\ point\ voltage\ (E_{rq}) < Required\ knee\ point\ voltage(V_k)$

$CT\ Tap\ is\ not\ adequate$

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Summary of Application : LineDifferentialProtection

CT-Protection Core

Summary							
Tap Details	**CT Primary current I_{pr} (A)**	**CT Secondary current I_{sr} (A)**	**Accuracy Class**	**Accuracy Limiting Factor (ALF)**	**Knee point voltage(V)**		**Results**
					Proposed	**Calculated**	
Tap-1	1250	1	5PR	20	340	495,9232	Not Adequate

Core ID: Core-3

Application Name: LineDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency 3-phase fault current for internal close-in fault current	$I_{k\,max}$	=	40000	A
	Maximum primary fundamental frequency 1-phase fault current for internal close-in fault current	$I_{ke\,max}$	=	40000	A
	Maximum primary fundamental frequency 3-phase fault current for external through fault current	$I_{t\,max}$	=	40000	A
	Maximum primary fundamental frequency 1-phase fault current for external through fault current	$I_{te\,max}$	=	40000	A
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Power Transformer included in the protection zone	YES/NO	=	No	

Core data

	Core Details		=	Core-3	
	Core Application		=	LineDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RED670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	75	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	1250	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A

·	CT Internal resistance	R_{ct}	=	7,5	Ohm
·	Rated output of the Current Transformer	S_n	=	30	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation ($\mathbf{R}_{lead}$):

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0C * (1 + (Max. conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max. conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((5{,}6065) * 75/1\right)}{1000}$$

$$R_{\text{lead}}\ = 0{,}841\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$$

$$I_{\text{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\text{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\text{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\text{relay}}\ = 0{,}02\ \ Ohm$$

Tap ID : 1 (LineDifferentialProtection)

CT knee point voltage Calculation as per Manual

(Ref: Annexure for relay manual)

$$Equivalent\ secondary\ e.m.f.for\ 3-ø\ internal\ close-in\ faults\ (E_{3-p\ in}) = I_{k\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + R_{lead} + R_{relay})$$

$$E_{3-p\ in} = 40000 * \left(\frac{1}{1250}\right) * (7{,}5 + 0{,}841 + 0{,}02)$$

$$E_{3-p\ in} = 267{,}552\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1-ø\ internal\ close-in\ faults\ (E_{1-p\ in}) = I_{ke\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + 2R_{lead} + R_{relay})$$

$$E_{1-p\ in} = 40000 * \left(\frac{1}{1250}\right) * (7{,}5 + (2 * 0{,}841) + 0{,}02)$$

$$E_{1-p\ in} = 267{,}552\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 3-ø\ external\ through\ faults\ (E_{3-p\ ex}) = 2 * I_{t\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + R_{lead} + R_{relay})$$

$$E_{3-p\ ex} = 2 * 40000 * \left(\frac{1}{1250}\right) * (7{,}5 + 0{,}841 + 0{,}02)$$

$$E_{3-p\ ex} = 535{,}104\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1-ø\ external\ through\ faults\ (E_{1-p\ ex}) = 2 * I_{te\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + 2R_{lead} + R_{relay})$$

$$E_{1-p\ ex} = 2 * 40000 * \left(\frac{1}{1250}\right) * (7{,}5 + (2 * 0{,}841) + 0{,}02)$$

$$E_{1-p\ ex} = 535{,}104\ V$$

$$Minimum\ required\ Knee\ point\ voltage\ (V_k) = Max(E_{3-p\ in}, E_{1-p\ in}, E_{3-p\ ex}, E_{1-p\ ex})$$

$$V_k = Max\ (267{,}552, \quad 267{,}552, \quad 535{,}104, 535{,}104)$$

$$V_k = 535{,}104\ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$CT\ secondary\ limiting\ e.m.f.\ E_{ALF} = ALF * I_{sr} * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$$

$$E_{ALF} = 20 * 1 * \left(7{,}5 + \frac{30}{1^2}\right)$$

$$E_{ALF} = 750\ V$$

$$E_{rq} = E_{ALF}$$

$$E_{rq} = 750\ V$$

$$Proposed\ knee\ point\ voltage\ (E_{rq}) > Required\ knee\ point\ voltage (V_k)$$

$CT\ Tap\ is\ adequate$

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Summary of Application : LineDifferentialProtection

CT-Protection Core

Summary							
Tap Details	***CT Primary current*** $I_{pr}\ (A)$	***CT Secondary current*** $I_{sr}\ (A)$	***Accuracy Class***	***Accuracy Limiting Factor*** (ALF)	***Knee point voltage*** (V)		***Results***
					Proposed	***Calculated***	
Tap-1	1250	1	5PR	20	750	535,104	Adequate

Core ID: Core-4

Application Name: LineDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency 3-phase fault current for internal close-in fault current	$I_{k\ max}$	=	40000	A
	Maximum primary fundamental frequency 1-phase fault current for internal close-in fault current	$I_{ke\ max}$	=	40000	A

	Maximum primary fundamental frequency 3-phase fault current for external through fault current	$I_{t\,max}$	=	40000	A
	Maximum primary fundamental frequency 1-phase fault current for external through fault current	$I_{te\,max}$	=	40000	A
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Power Transformer included in the protection zone	YES/NO	=	No	

Core data

	Core Details		=	Core-4	
	Core Application		=	LineDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RED670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	70	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	1600	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	7,5	Ohm
·	Rated output of the Current Transformer	S_n	=	10	VA
·	Accuracy limit factor	ALF	=	40	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$:
(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^{0}C * (1 + (Max.conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max.conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((5{,}6065) * 70/1\right)}{1000}$$

$$R_{\text{lead}}\ = 0{,}7849\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$$

$$I_{\text{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\text{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\text{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\text{relay}}\ = 0{,}02\ \ Ohm$$

Tap ID : 1 (LineDifferentialProtection)

CT knee point voltage Calculation as per Manual
(Ref: Annexure for relay manual)

$$Equivalent\ secondary\ e.m.f.for\ 3 - ø\ internal\ close - in\ faults\ (E_{3-p\ in}) = I_{k\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + R_{lead} + R_{relay}\right)$$

$$E_{3-p\ in} = 40000 * \left(\frac{1}{1600}\right) * (7{,}5 + 0{,}7849 + 0{,}02)$$

$$E_{3-p\ in} = 207{,}6225\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1 - ø\ internal\ close - in\ faults\ (E_{1-p\ in}) = I_{ke\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + 2R_{lead} + R_{relay}\right)$$

$$E_{1-p\ in} = 40000 * \left(\frac{1}{1600}\right) * (7{,}5 + (2 * 0{,}7849) + 0{,}02)$$

$$E_{1-p\ in} = 207{,}6225\ V$$

$Equivalent\ secondary\ e.m.f.for\ 3-ø\ external\ through\ faults\ (E_{3-p\ ex}) = 2 * I_{t\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + R_{lead} + R_{relay})$

$E_{3-p\ ex} = 2 * 40000 * \left(\frac{1}{1600}\right) * (7{,}5 + 0{,}7849 + 0{,}02)$

$E_{3-p\ ex} = 415{,}245\ V$

$Equivalent\ secondary\ e.m.f.for\ 1-ø\ external\ through\ faults\ (E_{1-p\ ex}) = 2 * I_{te\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + 2R_{lead} + R_{relay})$

$E_{1-p\ ex} = 2 * 40000 * \left(\frac{1}{1600}\right) * (7{,}5 + (2 * 0{,}7849) + 0{,}02)$

$E_{1-p\ ex} = 415{,}245\ V$

$Minimum\ required\ Knee\ point\ voltage\ (V_k) = Max(E_{3-p\ in}, E_{1-p\ in}, E_{3-p\ ex}, E_{1-p\ ex})$

$V_k = Max\ (207{,}6225, \quad 207{,}6225, \quad 415{,}245{,}415{,}245)$

$V_k = 415{,}245\ V$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$CT\ secondary\ limiting\ e.m.f.\ E_{ALF} = ALF * I_{sr} * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$

$E_{ALF} = 40 * 1 * \left(7{,}5 + \frac{10}{1^2}\right)$

$E_{ALF} = 700\ V$

$E_{rq} = E_{ALF}$

$E_{rq} = 700\ V$

$Proposed\ knee\ point\ voltage\ (E_{rq}) > Required\ knee\ point\ voltage(V_k)$

$CT\ Tap\ is\ adequate$

.

Summary of Application : LineDifferentialProtection

CT-Protection Core

Summary							
Tap Details	*CT Primary current* I_{pr} (A)	*CT Secondary* current I_{sr} (A)	*Accuracy Class*	*Accuracy Limiting Factor* (ALF)	*Knee point voltage*(V)		*Results*
					Proposed	*Calculated*	
Tap-1	1600	1	5PR	40	700	415,245	Adequate

Bay ID: Trafo Diff - 150(TD - 150)

Core ID: Core-1

Application Name: TransformerDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency phase to earth current that passed two main CTs without passing the t...	I_{tf}	=	6600	A
	Transformer rating	T_{MVA}	=	310	MVA
	Transformer impedance at principal tap	Z_t	=	18	%
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Type of System	$Solidly$ $/High\ impedance$	=	Solidly	

Core data

	Core Details		=	Core-1	
	Core Application		=	TransformerDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RET670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	65	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	1250	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	7	Ohm
·	Rated output of the Current Transformer	S_n	=	10	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$:

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0C * (1 + (Max.conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max.conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\mathrm{lead}} = \ 2 * \frac{\left((5{,}6065) * 65/1\right)}{1000}$$

$$R_{\mathrm{lead}}\ = 0{,}7288\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\mathrm{R}}) = I_{\mathrm{sr}}$$

$$I_{\mathrm{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\mathrm{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\mathrm{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\mathrm{relay}}\ = 0{,}02\ \ Ohm$$

Tap ID : 1 (TransformerDifferentialProtection)

Model:RET670, RET650

(Ref: ABB application manual, RET-670, section-2 requirements, Cl.no:2.1.6.1)

Fault Calculations

$$Rated\ primary\ current\ of\ the\ transformer\ (\mathrm{I_{nt}})\ = \frac{\mathrm{T_{MVA}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$$

$$\mathrm{I_{nt}}\ = \frac{310 * 1000000}{150 * \sqrt{3} * 1000}$$

$$\mathrm{I_{nt}} = 1193{,}1906\ \mathrm{A}$$

$$Through\ fault\ MVA\ of\ the\ Transformer\ \mathrm{T_{MVAe}} = \frac{\mathrm{T_{MVA}} * 100}{\mathrm{Z_t}}$$

$$\mathrm{T_{MVAe}} = \frac{310 * 100}{18}$$

$$\mathrm{T_{MVAe}} = 1722{,}2222\ MVA$$

$$Maximum\ primary\ fundamental\ frequency\ current\ that\ passes\ two\ main\ CTs\ and\ the\ transformer\ (\mathrm{I_{tf}}) = \frac{\mathrm{T_{MVAe}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = \frac{1722{,}2222 * 1000000}{150 * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = 6628{,}8363\ A$$

CT knee point voltage Calculation as per Manual

$$Equivalent\ secondary\ e.m.f.for\ close-in\ faults\ (E_{alreq1}) = 30 * I_{nt} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + \ 2R_{lead} + R_{relay}\right)$$

$$E_{alreq1} = 30 * 1193{,}1906 * \frac{1}{1250} * (7 + (2 * 0{,}7288) + 0{,}02)$$

$$E_{alreq1} = 242{,}7694\ V$$

$$Equivalent\ secondary\ e.m.f.for\ External\ faults\ (E_{alreq2}) = 2 * I_{tf} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + \ 2R_{lead} + R_{relay}\right)$$

$$E_{alreq2} = 2 * 6628{,}8363 * \frac{1}{1250} * (7 + (2 * \ 0{,}7288) + 0{,}02)$$

$$E_{alreq2} = 89{,}9146\ V$$

$$Maximum\ secondary\ limiting\ e.m.f\ (V_k) = Max(E_{alreq1}, E_{alreq2})$$

$$V_k = Max(242{,}7694{,}89{,}9146)$$

$$V_k = 242{,}7694\ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$Secondary\ limiting\ e.m.f.\ E_{ALF} = ALF\ * I_{sr}\ * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$$

$$E_{ALF} = 20\ * 1\ * \left(7 + \frac{10}{1^2}\right)$$

$$E_{ALF} = 340\ V$$

$$E_{rq} = E_{ALF}$$

$$E_{rq} = 340\ V$$

$$Proposed\ knee\ point\ voltage\ \left(E_{rq}\right) > Required\ knee\ point\ voltage(V_k)$$

$$CT\ Tap\ is\ adequate$$

.

Summary of Application : TransformerDifferentialProtection

CT-Protection Core

Summary

Tap Details	CT Primary current I_{pr} (A)	CT Secondary current I_{sr} (A)	Accuracy Class	Accuracy Limiting Factor (ALF)	Knee point voltage(V)		Results
					Proposed	Calculated	
Tap-1	1250	1	5PR	20	340	242,7694	Adequate

Core ID: Core-2

Application Name: TransformerDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency phase to earth current that passed two main CTs without passing the	I_{tf}	=	29000	A
	Transformer rating	T_{MVA}	=	325	MVA
	Transformer impedance at principal tap	Z_t	=	5	%
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Type of System	$Solidly$ $/High\ impedance$	=	Solidly	

Core data

	Core Details		=	Core-2	
	Core Application		=	TransformerDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RET670	

	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	65	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	1250	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	7	Ohm
·	Rated output of the Current Transformer	S_n	=	10	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$**:**

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0 C * (1 + (Max. conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4,61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5,6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max. conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((5,6065) * 65/1\right)}{1000}$$

$$R_{\text{lead}} = 0,7288\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\mathrm{R}}) = I_{\mathrm{sr}}$$

$$I_{\mathrm{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\mathrm{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\mathrm{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\mathrm{relay}}\ = 0{,}02\ \ Ohm$$

Tap ID : 1 (TransformerDifferentialProtection)

Model:RET670, RET650

(Ref: ABB application manual, RET-670, section-2 requirements, Cl.no:2.1.6.1)

Fault Calculations

$$Rated\ primary\ current\ of\ the\ transformer\ (\mathrm{I_{nt}})\ = \frac{\mathrm{T_{MVA}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$$

$$\mathrm{I_{nt}}\ = \frac{325 * 1000000}{150 * \sqrt{3} * 1000}$$

$$\mathrm{I_{nt}} = 1250{,}9256\ \mathrm{A}$$

$$Through\ fault\ MVA\ of\ the\ Transformer\ \mathrm{T_{MVAe}} = \frac{\mathrm{T_{MVA}} * 100}{\mathrm{Z_t}}$$

$$\mathrm{T_{MVAe}} = \frac{325 * 100}{5}$$

$$\mathrm{T_{MVAe}} = 6500\ MVA$$

$$Maximum\ primary\ fundamental\ frequency\ current\ that\ passes\ two\ main\ CTs\ and\ the\ transformer\ (\mathrm{I_{tf}}) = \frac{\mathrm{T_{MVAe}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = \frac{6500 * 1000000}{150 * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = 25018{,}5117\ A$$

CT knee point voltage Calculation as per Manual

$$Equivalent\ secondary\ e.m.f.for\ close-in\ faults\ (E_{alreq1}) = 30 * I_{nt} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$$

$$E_{alreq1} = 30 * 1250{,}9256 * \frac{1}{1250} * (7 + (2 * 0{,}7288) + 0{,}02)$$

$$E_{alreq1}\ = 254{,}5163\ V$$

$$Equivalent\ secondary\ e.m.f.for\ External\ faults\ (E_{\mathrm{alreq2}}) = 2 * I_{tf} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$$

$$E_{\mathrm{alreq2}}\ = 2 * 25018{,}5117 * \frac{1}{1250} * (7 + (2 *\ 0{,}7288) + 0{,}02)$$

$$E_{\mathrm{alreq2}} = 339{,}3551\ V$$

$Maximum\ secondary\ limiting\ e.m.f\ (V_k) = Max(E_{alreq1}, E_{alreq2})$

$V_k = Max(254{,}5163{,}339{,}3551)$

$V_k = 339{,}3551\ V$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$Secondary\ limiting\ e.m.f.\ E_{ALF} = ALF * I_{sr} * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$$

$$E_{ALF} = 20 * 1 * \left(7 + \frac{10}{1^2}\right)$$

$E_{ALF} = 340\ V$

$E_{rq} = E_{ALF}$

$E_{rq} = 340\ V$

$Proposed\ knee\ point\ voltage\ (E_{rq}) > Required\ knee\ point\ voltage(V_k)$

$CT\ Tap\ is\ adequate$

.

Summary of Application : TransformerDifferentialProtection

CT-Protection Core

Summary							
Tap Details	***CT Primary current I_{pr} (A)***	***CT Secondary current I_{sr} (A)***	***Accuracy Class***	***Accuracy Limiting Factor (ALF)***	***Knee point voltage(V)***		***Results***
					Proposed	***Calculated***	
Tap-1	1250	1	5PR	20	340	339,3551	Adequate

Core ID: Core-3

Application Name: TransformerDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency phase to earth current that passed two main CTs without passing the	I_{tf}	=	8000	A
	Transformer rating	T_{MVA}	=	310	MVA
	Transformer impedance at principal tap	Z_t	=	18	%
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Type of System	$Solidly$ $/High\ impedance$	=	Solidly	

Core data

	Core Details		=	Core-3	
	Core Application		=	TransformerDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RET670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	75	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	1250	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	7,5	Ohm
·	Rated output of the Current Transformer	S_n	=	30	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{\boldsymbol{lead}})$**:**

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^{0}C * (1 + (Max.conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max.conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = \ 2 * \frac{\left((5{,}6065) * 75/1\right)}{1000}$$

$$R_{\text{lead}}\ = 0{,}841\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$$

$$I_{\text{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\text{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\text{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\text{relay}}\ = 0{,}02\ \ Ohm$$

Tap ID : 1 (TransformerDifferentialProtection)

Model:RET670, RET650

(Ref: ABB application manual, RET-670, section-2 requirements, Cl.no:2.1.6.1)

Fault Calculations

$$Rated\ primary\ current\ of\ the\ transformer\ (\mathrm{I_{nt}})\ =\ \frac{\mathrm{T_{MVA}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$$

$$\mathrm{I_{nt}}\ =\ \frac{310 * 1000000}{150 * \sqrt{3} * 1000}$$

$$\mathrm{I_{nt}} = 1193{,}1906\ \mathrm{A}$$

$$Through\ fault\ MVA\ of\ the\ Transformer\ \mathrm{T_{MVAe}} = \frac{\mathrm{T_{MVA}} * 100}{\mathrm{Z_t}}$$

$$\mathrm{T_{MVAe}} = \frac{310 * 100}{18}$$

$$\mathrm{T_{MVAe}} = 1722{,}2222\ MVA$$

$$Maximum\ primary\ fundamental\ frequency\ current\ that\ passes\ two\ main\ CTs\ and\ the\ transformer\ (\mathrm{I_{tf}}) = \frac{\mathrm{T_{MVAe}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = \frac{1722{,}2222 * 1000000}{150 * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = 6628{,}8363\ A$$

CT knee point voltage Calculation as per Manual

$$Equivalent\ secondary\ e.m.f.for\ close - in\ faults\ (E_{alreq1}) = 30 * I_{nt} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$$

$$E_{alreq1} = 30 * 1193{,}1906 * \frac{1}{1250} * (7{,}5 + (2 * 0{,}841) + 0{,}02)$$

$$E_{alreq1} = 263{,}5138\ V$$

$$Equivalent\ secondary\ e.m.f.for\ External\ faults\ (E_{\mathrm{alreq2}}) = 2 * I_{tf} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$$

$$E_{\mathrm{alreq2}}\ = 2 * 6628{,}8363 * \frac{1}{1250} * (7{,}5 + (2 *\ 0{,}841) + 0{,}02)$$

$$E_{\mathrm{alreq2}} = 97{,}5977\ V$$

$$Maximum\ secondary\ limiting\ e.m.f\ (V_k) = Max(E_{alreq1}, E_{alreq2})$$

$$V_k = Max(263{,}5138{,}97{,}5977)$$

$$V_k = 263{,}5138\ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$Secondary\ limiting\ e.m.f.\ E_{ALF}\ = ALF\ * I_{sr}\ *\ \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$$

$$E_{ALF} = 20 * 1 * \left(7{,}5 + \frac{30}{1^2}\right)$$

$$E_{ALF} = 750\,V$$

$$E_{rq} = E_{ALF}$$

$$E_{rq} = 750\,V$$

$Proposed\ knee\ point\ voltage\ (E_{rq}) > Required\ knee\ point\ voltage(V_k)$

$CT\ Tap\ is\ adequate$

.

Summary of Application : TransformerDifferentialProtection

CT-Protection Core

Summary							
Tap Details	**CT Primary current I_{pr} (A)**	**CT Secondary current I_{sr} (A)**	**Accuracy Class**	**Accuracy Limiting Factor (ALF)**	**Knee point voltage(V)**		**Results**
					Proposed	**Calculated**	
Tap-1	1250	1	5PR	20	750	263,5138	Adequate

Core ID: Core-4

Application Name: TransformerDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency phase to earth current that passed two main CTs without passing the	I_{tf}	=	6000	A
	Transformer rating	T_{MVA}	=	200	MVA
	Transformer impedance at principal tap	Z_t	=	15	%
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Type of System	$Solidly$ $/High\ impedance$	=	Solidly	

Core data

	Core Details		=	Core-4	
	Core Application		=	TransformerDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RET670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	70	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	1600	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	7,5	Ohm
·	Rated output of the Current Transformer	S_n	=	10	VA

·	Accuracy limit factor	ALF	=	40	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$**:**

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^{0}C * (1 + (Max. conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max. conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((5{,}6065) * 70/1\right)}{1000}$$

$$R_{\text{lead}}\ = 0{,}7849\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$$

$$I_{\text{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\text{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\text{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\text{relay}}\ = 0{,}02\ \ Ohm$$

Tap ID : 1 (TransformerDifferentialProtection)

Model:RET670, RET650

(Ref: ABB application manual, RET-670, section-2 requirements, Cl.no:2.1.6.1)

Fault Calculations

$$Rated\ primary\ current\ of\ the\ transformer\ (\text{I}_{\text{nt}})\ = \frac{\text{T}_{\text{MVA}} * 1000000}{\text{U}_{\text{n}} * \sqrt{3} * 1000}$$

$$\text{I}_{\text{nt}}\ = \frac{200 * 1000000}{150 * \sqrt{3} * 1000}$$

$$\text{I}_{\text{nt}} = 769{,}8004\ \text{A}$$

$$Through\ fault\ MVA\ of\ the\ Transformer\ \mathrm{T_{MVAe}} = \frac{\mathrm{T_{MVA}} * 100}{\mathrm{Z_t}}$$

$$\mathrm{T_{MVAe}} = \frac{200 * 100}{15}$$

$$\mathrm{T_{MVAe}} = 1333{,}3333\ MVA$$

$$Maximum\ primary\ fundamental\ frequency\ current\ that\ passes\ two\ main\ CTs\ and\ the\ transformer\ (\mathrm{I_{tf}}) = \frac{\mathrm{T_{MVAe}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = \frac{1333{,}3333 * 1000000}{150 * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = 5132{,}0023\ A$$

CT knee point voltage Calculation as per Manual

$$Equivalent\ secondary\ e.m.f.for\ close - in\ faults\ (E_{alreq1}) = 30 * I_{nt} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + \ 2R_{lead} + R_{relay}\right)$$

$$E_{alreq1} = 30 * 769{,}8004 * \frac{1}{1600} * (7{,}5 + (2 * 0{,}7849) + 0{,}02)$$

$$E_{alreq1} = 131{,}2\ V$$

$$Equivalent\ secondary\ e.m.f.for\ External\ faults\ (E_{\mathrm{alreq2}}) = 2 * I_{tf} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + \ 2R_{lead} + R_{relay}\right)$$

$$E_{\mathrm{alreq2}} = 2 * 5132{,}0023 * \frac{1}{1600} * (7{,}5 + (2 * \ 0{,}7849) + 0{,}02)$$

$$E_{\mathrm{alreq2}} = 58{,}3111\ V$$

$$Maximum\ secondary\ limiting\ e.m.f\ (V_k) = Max(E_{alreq1}, E_{alreq2})$$

$$V_k = Max(131{,}2{,}58{,}3111)$$

$$V_k = 131{,}2\ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$Secondary\ limiting\ e.m.f.\ E_{ALF} = ALF\ * I_{sr}\ * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$$

$$E_{ALF} = 40\ * 1\ * \left(7{,}5 + \frac{10}{1^2}\right)$$

$$E_{ALF} = 700\ V$$

$$E_{rq} = E_{ALF}$$

$$E_{rq} = 700\ V$$

$$Proposed\ knee\ point\ voltage\ \left(E_{rq}\right) > Required\ knee\ point\ voltage(V_k)$$

$CT\ Tap\ is\ adequate$

.

Summary of Application : TransformerDifferentialProtection

CT-Protection Core

Summary							
$\boldsymbol{Tap\ Details}$	$\boldsymbol{CT\ Primary\ current}$ $\boldsymbol{I_{pr}\ (A)}$	$\boldsymbol{CT\ Secondary}$ **current** $\boldsymbol{I_{sr}\ (A)}$	$\boldsymbol{Accuracy\ Class}$	$\boldsymbol{Accuracy\ Limiting\ Factor}$ $\boldsymbol{(ALF)}$	$\boldsymbol{Knee\ point\ voltage (V)}$		$\boldsymbol{Results}$
					$\boldsymbol{Proposed}$	$\boldsymbol{Calculated}$	
Tap-1	1600	1	5PR	40	700	131,2	Adequate

Scenario 33kV

General data					
	Type of switchyard (AIS/GIS)		=	AIS	
	Nominal System voltage	U_n	=	33	kV
	System frequency	f_r	=	50	Hz
	Maximum Conductor temperature	°c	=	75	deg

Bay ID: Line Diff - 33(LD - 33)

Core ID: Core-1

Application Name: LineDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency 3-phase fault current for internal close-in fault current	$I_{k\,max}$	=	31500	A
	Maximum primary fundamental frequency 1-phase fault current for internal close-in fault current	$I_{ke\,max}$	=	31500	A
	Maximum primary fundamental frequency 3-phase fault current for external through fault current	$I_{t\,max}$	=	31500	A
	Maximum primary fundamental frequency 1-phase fault current for external through fault current	$I_{te\,max}$	=	31500	A
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Power Transformer included in the protection zone	YES/NO	=	No	

Core data

	Core Details		=	Core-1	
	Core Application		=	LineDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RED670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	1.5	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	12,1	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	13	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	2500	A

·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	14	Ohm
·	Rated output of the Current Transformer	S_n	=	15	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation ($\mathbf{R}_{lead}$):

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0 C * (1 + (Max.\,conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (12{,}1 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 14{,}7154\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max.\,conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((14{,}7154) * 13/1\right)}{1000}$$

$$R_{\text{lead}}\ = 0{,}3826\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$$

$$I_{\text{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\text{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\text{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\text{relay}}\ = 0{,}02\ \ Ohm$$

Tap ID : 1 (LineDifferentialProtection)

CT knee point voltage Calculation as per Manual

(Ref: Annexure for relay manual)

$$Equivalent\ secondary\ e.m.f.for\ 3-ø\ internal\ close-in\ faults\ (E_{3-p\ in}) = I_{k\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + R_{lead} + R_{relay})$$

$$E_{3-p\ in} = 31500 * \left(\frac{1}{2500}\right) * (14 + 0{,}3826 + 0{,}02)$$

$$E_{3-p\ in} = 181{,}4728\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1-ø\ internal\ close-in\ faults\ (E_{1-p\ in}) = I_{ke\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + 2R_{lead} + R_{relay})$$

$$E_{1-p\ in} = 31500 * \left(\frac{1}{2500}\right) * (14 + (2 * 0{,}3826) + 0{,}02)$$

$$E_{1-p\ in} = 181{,}4728\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 3-ø\ external\ through\ faults\ (E_{3-p\ ex}) = 2 * I_{t\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + R_{lead} + R_{relay})$$

$$E_{3-p\ ex} = 2 * 31500 * \left(\frac{1}{2500}\right) * (14 + 0{,}3826 + 0{,}02)$$

$$E_{3-p\ ex} = 362{,}9455\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1-ø\ external\ through\ faults\ (E_{1-p\ ex}) = 2 * I_{te\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + 2R_{lead} + R_{relay})$$

$$E_{1-p\ ex} = 2 * 31500 * \left(\frac{1}{2500}\right) * (14 + (2 * 0{,}3826) + 0{,}02)$$

$$E_{1-p\ ex} = 362{,}9455\ V$$

$$Minimum\ required\ Knee\ point\ voltage\ (V_k) = Max(E_{3-p\ in}, E_{1-p\ in}, E_{3-p\ ex}, E_{1-p\ ex})$$

$$V_k = Max\ (181{,}4728, \quad 181{,}4728, \quad 362{,}9455{,}362{,}9455)$$

$$V_k = 362{,}9455\ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$CT\ secondary\ limiting\ e.m.f.\ E_{ALF} = ALF * I_{sr} * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$$

$$E_{ALF} = 20 * 1 * \left(14 + \frac{15}{1^2}\right)$$

$$E_{ALF} = 580\ V$$

$$E_{rq} = E_{ALF}$$

$$E_{rq} = 580\ V$$

$$Proposed\ knee\ point\ voltage\ (E_{rq}) > Required\ knee\ point\ voltage(V_k)$$

CT Tap is adequate

.

Summary of Application : LineDifferentialProtection

CT-Protection Core

Summary							
Tap Details	***CT Primary current*** $I_{pr}\ (A)$	***CT Secondary*** **current** $I_{sr}\ (A)$	***Accuracy Class***	***Accuracy Limiting Factor*** (ALF)	***Knee point voltage*** (V)		***Results***
					Proposed	***Calculated***	
Tap-1	2500	1	5PR	20	580	362,9455	Adequate

Core ID: Core-2

Application Name: LineDifferentialProtection

Relay Input data					
	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency 3-phase fault current for internal close-in fault current	$I_{k\ max}$	=	40000	A
	Maximum primary fundamental frequency 1-phase fault current for internal close-in fault current	$I_{ke\ max}$	=	40000	A

	Maximum primary fundamental frequency 3-phase fault current for external through fault current	$I_{t\,max}$	=	40000	A
	Maximum primary fundamental frequency 1-phase fault current for external through fault current	$I_{te\,max}$	=	40000	A
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Power Transformer included in the protection zone	YES/NO	=	No	

Core data

	Core Details		=	Core-2	
	Core Application		=	LineDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RED670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	110	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	3000	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	20	Ohm
·	Rated output of the Current Transformer	S_n	=	30	VA
·	Accuracy limit factor	ALF	=	60	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation ($\mathbf{R}_{lead}$):
(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0C * (1 + (Max. conductor\ temp. -20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max. conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((5{,}6065) * 110/1\right)}{1000}$$

$$R_{\text{lead}}\ = 1{,}2334\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$$

$$I_{\text{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\text{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\text{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\text{relay}}\ = 0{,}02\ \ Ohm$$

Tap ID : 1 (LineDifferentialProtection)

CT knee point voltage Calculation as per Manual
(Ref: Annexure for relay manual)

$$Equivalent\ secondary\ e.m.f.for\ 3 - ø\ internal\ close - in\ faults\ (E_{3-p\ in}) =\ I_{k\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ R_{lead} + R_{relay}\right)$$

$$E_{3-p\ in} = 40000 * \left(\frac{1}{3000}\right) *\ (20 +\ 1{,}2334 +\ 0{,}02)$$

$$E_{3-p\ in} =\ 283{,}3787\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1 - ø\ internal\ close - in\ faults\ (E_{1-p\ in}) =\ I_{ke\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$$

$$E_{1-p\ in} = 40000 * \left(\frac{1}{3000}\right) *\ (20 +\ (2 *\ 1{,}2334) +\ 0{,}02)$$

$$E_{1-p\ in} =\ 283{,}3787\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 3-ø\ external\ through\ faults\ (E_{3-p\ ex}) = 2 * I_{t\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + R_{lead} + R_{relay})$$

$$E_{3-p\ ex} = 2 * 40000 * \left(\frac{1}{3000}\right) * (20 + 1{,}2334 + 0{,}02)$$

$$E_{3-p\ ex} = 566{,}7573\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1-ø\ external\ through\ faults\ (E_{1-p\ ex}) = 2 * I_{te\ max} * \frac{I_{sr}}{I_{pr}} * (R_{ct} + 2R_{lead} + R_{relay})$$

$$E_{1-p\ ex} = 2 * 40000 * \left(\frac{1}{3000}\right) * (20 + (2 * 1{,}2334) + 0{,}02)$$

$$E_{1-p\ ex} = 566{,}7573\ V$$

$$Minimum\ required\ Knee\ point\ voltage\ (V_k) = Max(E_{3-p\ in}, E_{1-p\ in}, E_{3-p\ ex}, E_{1-p\ ex})$$

$$V_k = Max\ (283{,}3787, \quad 283{,}3787, \quad 566{,}7573, 566{,}7573)$$

$$V_k = 566{,}7573\ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$CT\ secondary\ limiting\ e.m.f.\ E_{ALF} = ALF * I_{sr} * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$$

$$E_{ALF} = 60 * 1 * \left(20 + \frac{30}{1^2}\right)$$

$$E_{ALF} = 3000\ V$$

$$E_{rq} = E_{ALF}$$

$$E_{rq} = 3000\ V$$

$$Proposed\ knee\ point\ voltage\ (E_{rq}) > Required\ knee\ point\ voltage(V_k)$$

CT Tap is adequate

.

Summary of Application : LineDifferentialProtection

CT-Protection Core

Summary							
Tap Details	***CT Primary current*** I_{pr} (A)	***CT Secondary current*** I_{sr} (A)	***Accuracy Class***	***Accuracy Limiting Factor*** (ALF)	***Knee point voltage*** (V)		***Results***
					Proposed	***Calculated***	
Tap-1	3000	1	5PR	60	3000	566,7573	Adequate

Core ID: Core-3

Application Name: LineDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency 3-phase fault current for internal close-in fault current	$I_{k\,max}$	=	29000	A
	Maximum primary fundamental frequency 1-phase fault current for internal close-in fault current	$I_{ke\,max}$	=	29000	A
	Maximum primary fundamental frequency 3-phase fault current for external through fault current	$I_{t\,max}$	=	30000	A
	Maximum primary fundamental frequency 1-phase fault current for external through fault current	$I_{te\,max}$	=	30000	A
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Power Transformer included in the protection zone	YES/NO	=	No	

Core data

	Core Details		=	Core-3	

	Core Application		=	LineDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RED670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	55	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	2500	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	14	Ohm
·	Rated output of the Current Transformer	S_n	=	20	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$**:**

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0C * (1 + (Max.\ conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max.conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\mathrm{lead}} = 2 * \frac{\left((5,6065) * 55/1\right)}{1000}$$

$$R_{\mathrm{lead}} = 0,6167\ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\mathrm{R}}) = I_{\mathrm{sr}}$$

$$I_{\mathrm{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\mathrm{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\mathrm{relay}} = \frac{0,02}{1^2}$$

$$R_{\mathrm{relay}} = 0,02\ Ohm$$

Tap ID : 1 (LineDifferentialProtection)

CT knee point voltage Calculation as per Manual

(Ref: Annexure for relay manual)

$$Equivalent\ secondary\ e.m.f.for\ 3-ø\ internal\ close-in\ faults\ (E_{3-p\ in}) = I_{k\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + R_{lead} + R_{relay}\right)$$

$$E_{3-p\ in} = 29000 * \left(\frac{1}{2500}\right) * (14 + 0,6167 + 0,02)$$

$$E_{3-p\ in} = 169,7857\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1-ø\ internal\ close-in\ faults\ (E_{1-p\ in}) = I_{ke\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + 2R_{lead} + R_{relay}\right)$$

$$E_{1-p\ in} = 29000 * \left(\frac{1}{2500}\right) * (14 + (2 * 0,6167) + 0,02)$$

$$E_{1-p\ in} = 169,7857\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 3-ø\ external\ through\ faults\ (E_{3-p\ ex}) = 2 * I_{t\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + R_{lead} + R_{relay}\right)$$

$$E_{3-p\ ex} = 2 * 30000 * \left(\frac{1}{2500}\right) * (14 + 0,6167 + 0,02)$$

$$E_{3-p\ ex} = 351,2808\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1-ø\ external\ through\ faults\ (E_{1-p\ ex}) = 2 * I_{te\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + 2R_{lead} + R_{relay}\right)$$

$$E_{1-p\ ex} = 2 * 30000 * \left(\frac{1}{2500}\right) * (14 + (2 * 0,6167) + 0,02)$$

$$E_{1-p\ ex} = 351,2808\ V$$

$Minimum\ required\ Knee\ point\ voltage\ (V_k) =\ Max(E_{3-p\ in}, E_{1-p\ in}, E_{3-p\ ex}, E_{1-p\ ex})$

$V_k = Max\ (169{,}7857,\quad 169{,}7857,\quad 351{,}2808{,}351{,}2808)$

$V_k = 351{,}2808\ V$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$CT\ secondary\ limiting\ e.m.f.\ E_{ALF} = ALF\ * I_{sr}\ * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$

$E_{ALF} = 20\ * 1\ * \left(14 + \frac{20}{1^2}\right)$

$E_{ALF} = 680\ V$

$E_{rq} = E_{ALF}$

$E_{rq} = 680\ V$

$Proposed\ knee\ point\ voltage\ (E_{rq}) > Required\ knee\ point\ voltage(V_k)$

$CT\ Tap\ is\ adequate$

.

Summary of Application : LineDifferentialProtection

CT-Protection Core

Summary							
Tap Details	***CT Primary current*** $I_{pr}\ (A)$	***CT Secondary current*** $I_{sr}\ (A)$	***Accuracy Class***	***Accuracy Limiting Factor*** (ALF)	***Knee point voltage***(V)		***Results***
					Proposed	***Calculated***	
Tap-1	2500	1	5PR	20	680	351,2808	Adequate

Core ID: Core-4

Application Name: LineDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency 3-phase fault current for internal close-in fault current	$I_{k\,max}$	=	29000	A
	Maximum primary fundamental frequency 1-phase fault current for internal close-in fault current	$I_{ke\,max}$	=	29000	A
	Maximum primary fundamental frequency 3-phase fault current for external through fault current	$I_{t\,max}$	=	29000	A
	Maximum primary fundamental frequency 1-phase fault current for external through fault current	$I_{te\,max}$	=	29000	A
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Power Transformer included in the protection zone	YES/NO	=	No	

Core data

	Core Details		=	Core-4	
	Core Application		=	LineDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RED670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$

	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	80	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	3000	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	18	Ohm
·	Rated output of the Current Transformer	S_n	=	20	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$:

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0 C * (1 + (Max.conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max.conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((5{,}6065) * 80/1\right)}{1000}$$

$$R_{\text{lead}} = 0{,}897\ \ Ohm$$

Relay Parameters Calculation

$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$

$I_{\mathrm{R}} = 1\,A$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\mathrm{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\mathrm{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\mathrm{relay}} = 0{,}02\ \ Ohm$$

Tap ID : 1 (LineDifferentialProtection)

CT knee point voltage Calculation as per Manual
(Ref: Annexure for relay manual)

$$Equivalent\ secondary\ e.m.f.for\ 3-ø\ internal\ close-in\ faults\ (E_{3-p\ in}) = I_{k\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + R_{lead} + R_{relay}\right)$$

$$E_{3-p\ in} = 29000 * \left(\frac{1}{3000}\right) * (18 + 0{,}897 + 0{,}02)$$

$$E_{3-p\ in} = 182{,}8643\,V$$

$$Equivalent\ secondary\ e.m.f.for\ 1-ø\ internal\ close-in\ faults\ (E_{1-p\ in}) = I_{ke\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + 2R_{lead} + R_{relay}\right)$$

$$E_{1-p\ in} = 29000 * \left(\frac{1}{3000}\right) * (18 + (2 * 0{,}897) + 0{,}02)$$

$$E_{1-p\ in} = 182{,}8643\,V$$

$$Equivalent\ secondary\ e.m.f.for\ 3-ø\ external\ through\ faults\ (E_{3-p\ ex}) = 2 * I_{t\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + R_{lead} + R_{relay}\right)$$

$$E_{3-p\ ex} = 2 * 29000 * \left(\frac{1}{3000}\right) * (18 + 0{,}897 + 0{,}02)$$

$$E_{3-p\ ex} = 365{,}7287\,V$$

$$Equivalent\ secondary\ e.m.f.for\ 1-ø\ external\ through\ faults\ (E_{1-p\ ex}) = 2 * I_{te\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + 2R_{lead} + R_{relay}\right)$$

$$E_{1-p\ ex} = 2 * 29000 * \left(\frac{1}{3000}\right) * (18 + (2 * 0{,}897) + 0{,}02)$$

$$E_{1-p\ ex} = 365{,}7287\,V$$

$$Minimum\ required\ Knee\ point\ voltage\ (V_k) = Max(E_{3-p\ in}, E_{1-p\ in}, E_{3-p\ ex}, E_{1-p\ ex})$$

$$V_k = Max\,(182{,}8643, \quad 182{,}8643, \quad 365{,}7287, 365{,}7287)$$

$$V_k = 365{,}7287\ \ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$CT\ secondary\ limiting\ e.m.f.\ E_{ALF} = ALF * I_{sr} * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$$

$$E_{ALF} = 20 * 1 * \left(18 + \frac{20}{1^2}\right)$$

$$E_{ALF} = 760\,V$$

$$E_{rq} = E_{ALF}$$

$$E_{rq} = 760\,V$$

$Proposed\ knee\ point\ voltage\ \left(E_{rq}\right) > Required\ knee\ point\ voltage(V_k)$

$CT\ Tap\ is\ adequate$

.

Summary of Application : LineDifferentialProtection

CT-Protection Core

Summary							
Tap Details	**CT Primary current I_{pr} (A)**	**CT Secondary current I_{sr} (A)**	**Accuracy Class**	**Accuracy Limiting Factor (ALF)**	**Knee point voltage(V)**		**Results**
					Proposed	**Calculated**	
Tap-1	3000	1	5PR	20	760	365,7287	Adequate

Core ID: Core-5

Application Name: LineDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency 3-phase fault current for internal close-in fault current	$I_{k\,max}$	=	8400	A
	Maximum primary fundamental frequency 1-phase fault current for internal close-in fault current	$I_{ke\,max}$	=	8400	A
	Maximum primary fundamental frequency 3-phase fault current for external through fault current	$I_{t\,max}$	=	8400	A
	Maximum primary fundamental frequency 1-phase fault current for external through fault current	$I_{te\,max}$	=	8400	A
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Power Transformer included in the protection zone	YES/NO	=	No	

Core data

	Core Details		=	Core-5	
	Core Application		=	LineDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RED670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	2.5	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	7,41	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	40	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	500	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	2	Ohm
·	Rated output of the Current Transformer	S_n	=	10	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$**:**

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0 C * (1 + (Max.\, conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (7{,}41 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 9{,}0117\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max.\, conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((9{,}0117) * 40/1\right)}{1000}$$

$$R_{\text{lead}}\ = 0{,}7209\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$$

$$I_{\text{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\text{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\text{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\text{relay}}\ = 0{,}02\ \ Ohm$$

Tap ID : 1 (LineDifferentialProtection)

CT knee point voltage Calculation as per Manual

(Ref: Annexure for relay manual)

$$Equivalent\ secondary\ e.m.f.for\ 3-ø\ internal\ close-in\ faults\ (E_{3-p\ in}) = I_{k\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + R_{lead} + R_{relay}\right)$$

$$E_{3-p\ in} = 8400 * \left(\frac{1}{500}\right) * (2 + 0{,}7209 + 0{,}02)$$

$$E_{3-p\ in} = 46{,}0471\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1-ø\ internal\ close-in\ faults\ (E_{1-p\ in}) = I_{ke\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + 2R_{lead} + R_{relay}\right)$$

$$E_{1-p\ in} = 8400 * \left(\frac{1}{500}\right) * (2 + (2 * 0{,}7209) + 0{,}02)$$

$$E_{1-p\ in} = 46{,}0471\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 3-ø\ external\ through\ faults\ (E_{3-p\ ex}) = 2 * I_{t\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + R_{lead} + R_{relay}\right)$$

$$E_{3-p\ ex} = 2 * 8400 * \left(\frac{1}{500}\right) * (2 + 0{,}7209 + 0{,}02)$$

$$E_{3-p\ ex} = 92{,}0942\ V$$

$$Equivalent\ secondary\ e.m.f.for\ 1-ø\ external\ through\ faults\ (E_{1-p\ ex}) = 2 * I_{te\ max} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + 2R_{lead} + R_{relay}\right)$$

$$E_{1-p\ ex} = 2 * 8400 * \left(\frac{1}{500}\right) * (2 + (2 * 0{,}7209) + 0{,}02)$$

$$E_{1-p\ ex} = 92{,}0942\ V$$

$$Minimum\ required\ Knee\ point\ voltage\ (V_k) = Max(E_{3-p\ in}, E_{1-p\ in}, E_{3-p\ ex}, E_{1-p\ ex})$$

$$V_k = Max\ (46{,}0471, \quad 46{,}0471, \quad 92{,}0942, 92{,}0942)$$

$$V_k = 92{,}0942\ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$CT\ secondary\ limiting\ e.m.f.\ E_{ALF} = ALF * I_{sr} * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$$

$$E_{ALF} = 20 * 1 * \left(2 + \frac{10}{1^2}\right)$$

$$E_{ALF} = 240\ V$$

$$E_{rq} = E_{ALF}$$

$$E_{rq} = 240\ V$$

$Proposed\ knee\ point\ voltage\ (E_{rq}) > Required\ knee\ point\ voltage(V_k)$

$CT\ Tap\ is\ adequate$

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Summary of Application : LineDifferentialProtection

CT-Protection Core

Summary							
$Tap\ Details$	$CT\ Primary$ current $I_{pr}\ (A)$	$CT\ Secondary$ current $I_{sr}\ (A)$	$Accuracy\ Class$	$Accuracy\ Limiting\ Factor\ (ALF)$	$Knee\ point\ voltage(V)$		$Results$
					$Proposed$	$Calculated$	
Tap-1	500	1	5PR	20	240	92,0942	Adequate

Bay ID: Trafo Diff - 33(TD - 33)

Core ID: Core-1

Application Name: TransformerDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency phase to earth current that passed two main CTs without passing the	I_{tf}	=	16000	A
	Transformer rating	T_{MVA}	=	110	MVA
	Transformer impedance at principal tap	Z_t	=	12	%
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Type of System	$Solidly$ $/High\ impedance$	=	Solidly	

Core data

	Core Details		=	Core-1	
	Core Application		=	TransformerDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RET670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	1.5	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	12,1	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	13	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	2500	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	14	Ohm
·	Rated output of the Current Transformer	S_n	=	15	VA

·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$**:**

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0 C * (1 + (Max. conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (12{,}1 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 14{,}7154\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max. conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((14{,}7154) * 13/1\right)}{1000}$$

$$R_{\text{lead}}\ = 0{,}3826\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$$

$$I_{\text{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\text{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\text{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\text{relay}}\ = 0{,}02\ \ Ohm$$

Tap ID : 1 (TransformerDifferentialProtection)

Model:RET670, RET650

(Ref: ABB application manual, RET-670, section-2 requirements, Cl.no:2.1.6.1)

Fault Calculations

$$Rated\ primary\ current\ of\ the\ transformer\ (\text{I}_{\text{nt}})\ = \frac{\text{T}_{\text{MVA}} * 1000000}{\text{U}_{\text{n}} * \sqrt{3} * 1000}$$

$$\text{I}_{\text{nt}}\ = \frac{110 * 1000000}{33 * \sqrt{3} * 1000}$$

$$\text{I}_{\text{nt}} = 1924{,}5009\ \text{A}$$

$$Through\ fault\ MVA\ of\ the\ Transformer\ \mathrm{T_{MVAe}} = \frac{\mathrm{T_{MVA}} * 100}{\mathrm{Z_t}}$$

$$\mathrm{T_{MVAe}} = \frac{110 * 100}{12}$$

$$\mathrm{T_{MVAe}} = 916{,}6667\ MVA$$

$$Maximum\ primary\ fundamental\ frequency\ current\ that\ passes\ two\ main\ CTs\ and\ the\ transformer\ (\mathrm{I_{tf}}) = \frac{\mathrm{T_{MVAe}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = \frac{916{,}6667 * 1000000}{33 * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = 16037{,}5081\ A$$

CT knee point voltage Calculation as per Manual

$$Equivalent\ secondary\ e.m.f.for\ close - in\ faults\ (E_{alreq1}) = 30 * I_{nt} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$$

$$E_{alreq1} = 30 * 1924{,}5009 * \frac{1}{2500} * (14 + (2 * 0{,}3826) + 0{,}02)$$

$$E_{alreq1} = 341{,}4496\ V$$

$$Equivalent\ secondary\ e.m.f.for\ External\ faults\ (E_{\mathrm{alreq2}}) = 2 * I_{tf} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$$

$$E_{\mathrm{alreq2}}\ = 2 * 16037{,}5081 * \frac{1}{2500} * (14 + (2 *\ 0{,}3826) + 0{,}02)$$

$$E_{\mathrm{alreq2}} = 189{,}6942\ V$$

$$Maximum\ secondary\ limiting\ e.m.f\ (V_k) = Max(E_{alreq1}, E_{alreq2})$$

$$V_k = Max(341{,}4496, 189{,}6942)$$

$$V_k = 341{,}4496\ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$Secondary\ limiting\ e.m.f.\ E_{ALF}\ = ALF\ * I_{sr}\ * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$$

$$E_{ALF}\ = 20\ * 1\ * \left(14 + \frac{15}{1^2}\right)$$

$$E_{ALF}\ = 580\ V$$

$$E_{rq}\ = E_{ALF}$$

$$E_{rq}\ = 580\ V$$

$$Proposed\ knee\ point\ voltage\ \left(E_{rq}\right) > Required\ knee\ point\ voltage(V_k)$$

$CT\ Tap\ is\ adequate$

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Summary of Application : TransformerDifferentialProtection

CT-Protection Core

Summary							
Tap Details	***CT Primary current*** $I_{pr}\ (A)$	***CT Secondary*** **current** $I_{sr}\ (A)$	***Accuracy Class***	***Accuracy Limiting Factor*** (ALF)	***Knee point voltage***(V)		***Results***
					Proposed	***Calculated***	
Tap-1	2500	1	5PR	20	580	341,4496	Adequate

Core ID: Core-2

Application Name: TransformerDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency phase to earth current that passed two main CTs without passing the	I_{tf}	=	29000	A
	Transformer rating	T_{MVA}	=	175	MVA
	Transformer impedance at principal tap	Z_t	=	12	%

	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Type of System	*Solidly /High impedance*	=	Solidly	

Core data

	Core Details		=	Core-2	
	Core Application		=	TransformerDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RET670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	110	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	3000	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	20	Ohm
·	Rated output of the Current Transformer	S_n	=	30	VA
·	Accuracy limit factor	ALF	=	60	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$:

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0 C * (1 + (Max.conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max.conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((5{,}6065) * 110/1\right)}{1000}$$

$$R_{\text{lead}}\ = 1{,}2334\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$$

$$I_{\text{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\text{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\text{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\text{relay}}\ = 0{,}02\ \ Ohm$$

Tap ID : 1 (TransformerDifferentialProtection)

Model:RET670, RET650

(Ref: ABB application manual, RET-670, section-2 requirements, Cl.no:2.1.6.1)

Fault Calculations

$$Rated\ primary\ current\ of\ the\ transformer\ (\text{I}_{\text{nt}})\ = \frac{\text{T}_{\text{MVA}} * 1000000}{\text{U}_{\text{n}} * \sqrt{3} * 1000}$$

$$\text{I}_{\text{nt}}\ = \frac{175 * 1000000}{33 * \sqrt{3} * 1000}$$

$$\text{I}_{\text{nt}} = 3061{,}706\ \text{A}$$

$$Through\ fault\ MVA\ of\ the\ Transformer\ \text{T}_{\text{MVAe}} = \frac{\text{T}_{\text{MVA}} * 100}{\text{Z}_{\text{t}}}$$

$$\text{T}_{\text{MVAe}} = \frac{175 * 100}{12}$$

$$\text{T}_{\text{MVAe}} = 1458{,}3333\ MVA$$

$$Maximum\ primary\ fundamental\ frequency\ current\ that\ passes\ two\ main\ CTs\ and\ the\ transformer\ (\text{I}_{\text{tf}}) = \frac{\text{T}_{\text{MVAe}} * 1000000}{\text{U}_{\text{n}} * \sqrt{3} * 1000}$$

$$\text{I}_{\text{tf}} = \frac{1458{,}3333 * 1000000}{33 * \sqrt{3} * 1000}$$

$$\text{I}_{\text{tf}} = 25514{,}2159\ A$$

CT knee point voltage Calculation as per Manual

$$Equivalent\ secondary\ e.m.f.for\ close-in\ faults\ (E_{alreq1}) = 30 * I_{nt} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$$

$$E_{alreq1} = 30 * 3061{,}706 * \frac{1}{3000} * (20 + (2 * 1{,}2334) + 0{,}02)$$

$$E_{alreq1} = 688{,}4797\ V$$

$$Equivalent\ secondary\ e.m.f.for\ External\ faults\ (E_{alreq2}) = 2 * I_{tf} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$$

$$E_{alreq2}\ = 2 * 25514{,}2159 * \frac{1}{3000} * (20 + (2 *\ 1{,}2334) + 0{,}02)$$

$$E_{alreq2} = 382{,}4887\ V$$

$$Maximum\ secondary\ limiting\ e.m.f\ (V_k) = Max(E_{alreq1}, E_{alreq2})$$

$$V_k = Max(688{,}4797{,}382{,}4887)$$

$$V_k = 688{,}4797\ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$Secondary\ limiting\ e.m.f.\ \ E_{ALF}\ = ALF\ \ * I_{sr}\ \ *\ \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$$

$$E_{ALF}\ = 60\ \ * 1\ \ *\ \left(20 + \frac{30}{1^2}\right)$$

$$E_{ALF}\ = 3000\ V$$

$$E_{rq}\ = E_{ALF}$$

$$E_{rq}\ = 3000\ V$$

$$Proposed\ knee\ point\ voltage\ \left(E_{rq}\right) > Required\ knee\ point\ voltage(V_k)$$

$$CT\ Tap\ is\ adequate$$

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Summary of Application : TransformerDifferentialProtection

CT-Protection Core

Summary							
Tap Details	***CT Primary current I_{pr} (A)***	***CT Secondary current I_{sr} (A)***	***Accuracy Class***	***Accuracy Limiting Factor (ALF)***	***Knee point voltage(V)***		***Results***
					Proposed	***Calculated***	
Tap-1	3000	1	5PR	60	3000	688,4797	Adequate

Core ID: Core-3

Application Name: TransformerDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency phase to earth current that passed two main CTs without passing the	I_{tf}	=	25000	A
	Transformer rating	T_{MVA}	=	155	MVA
	Transformer impedance at principal tap	Z_t	=	12	%
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Type of System	$Solidly$ $/High\ impedance$	=	Solidly	

Core data

	Core Details		=	Core-3	
	Core Application		=	TransformerDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos

	Relay Make		=	ABB	
	Relay Model		=	RET670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	55	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	2500	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	14	Ohm
·	Rated output of the Current Transformer	S_n	=	20	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$**:**

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0 C * (1 + (Max. conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max.conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((5{,}6065) * 55/1\right)}{1000}$$

$$R_{\text{lead}} = 0{,}6167\ \ Ohm$$

Relay Parameters Calculation

$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\mathrm{R}}) = I_{\mathrm{sr}}$

$I_{\mathrm{R}} = 1\ A$

$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\mathrm{relay}}) = \frac{S_R}{I_R^2}$

$R_{\mathrm{relay}} = \frac{0{,}02}{1^2}$

$R_{\mathrm{relay}}\ = 0{,}02\ \ Ohm$

Tap ID : 1 (TransformerDifferentialProtection)

Model:RET670, RET650

(Ref: ABB application manual, RET-670, section-2 requirements, Cl.no:2.1.6.1)

Fault Calculations

$Rated\ primary\ current\ of\ the\ transformer\ (\mathrm{I_{nt}})\ = \frac{\mathrm{T_{MVA}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$

$\mathrm{I_{nt}}\ = \frac{155 * 1000000}{33 * \sqrt{3} * 1000}$

$\mathrm{I_{nt}} = 2711{,}7967\ \mathrm{A}$

$Through\ fault\ MVA\ of\ the\ Transformer\ \mathrm{T_{MVAe}} = \frac{\mathrm{T_{MVA}} * 100}{\mathrm{Z_t}}$

$\mathrm{T_{MVAe}} = \frac{155 * 100}{12}$

$\mathrm{T_{MVAe}} = 1291{,}6667\ MVA$

$Maximum\ primary\ fundamental\ frequency\ current\ that\ passes\ two\ main\ CTs\ and\ the\ transformer\ (\mathrm{I_{tf}}) = \frac{\mathrm{T_{MVAe}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$

$\mathrm{I_{tf}} = \frac{1291{,}6667 * 1000000}{33 * \sqrt{3} * 1000}$

$\mathrm{I_{tf}} = 22598{,}3066\ A$

CT knee point voltage Calculation as per Manual

$Equivalent\ secondary\ e.m.f.for\ close-in\ faults\ (E_{alreq1}) = 30 * I_{nt} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$

$E_{alreq1} = 30 * 2711{,}7967 * \frac{1}{2500} * (14 + (2 * 0{,}6167) + 0{,}02)$

$E_{alreq1}\ = 496{,}3694\ V$

$Equivalent\ secondary\ e.m.f.for\ External\ faults\ (E_{\mathrm{alreq2}}) = 2 * I_{tf} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$

$E_{alreq2}\ = 2 * 22598{,}3066 * \frac{1}{2500} * (14 + (2 *\ 0{,}6167) + 0{,}02)$

$E_{alreq2} = 275{,}7608\ V$

$Maximum\ secondary\ limiting\ e.m.f\ (V_k) = Max(E_{alreq1}, E_{alreq2})$

$V_k = Max(496{,}3694{,}275{,}7608)$

$V_k = 496{,}3694\ V$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$Secondary\ limiting\ e.m.f.\ E_{ALF} = ALF * I_{sr} * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$

$E_{ALF} = 20 * 1 * \left(14 + \frac{20}{1^2}\right)$

$E_{ALF} = 680\ V$

$E_{rq} = E_{ALF}$

$E_{rq} = 680\ V$

$Proposed\ knee\ point\ voltage\ \left(E_{rq}\right) > Required\ knee\ point\ voltage(V_k)$

$CT\ Tap\ is\ adequate$

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Summary of Application : TransformerDifferentialProtection

CT-Protection Core

Summary							
Tap Details	***CT Primary current I_{pr} (A)***	***CT Secondary current I_{sr} (A)***	***Accuracy Class***	***Accuracy Limiting Factor (ALF)***	***Knee point voltage(V)***		***Results***
					Proposed	***Calculated***	
Tap-1	2500	1	5PR	20	680	496,3694	Adequate

Core ID: Core-4

Application Name: TransformerDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency phase to earth current that passed two main CTs without passing the t...	I_{tf}	=	25000	A
	Transformer rating	T_{MVA}	=	155	MVA
	Transformer impedance at principal tap	Z_t	=	10	%
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Type of System	$Solidly$ $/High\ impedance$	=	Solidly	

Core data

	Core Details		=	Core-4	
	Core Application		=	TransformerDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RET670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	4	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	4,61	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	80	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	3000	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A
·	CT Internal resistance	R_{ct}	=	18	Ohm
·	Rated output of the Current Transformer	S_n	=	20	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation ($\mathbf{R}_{lead}$):

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0 C * (1 + (Max. conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (4{,}61 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 5{,}6065\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max. conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = 2 * \frac{\left((5{,}6065) * 80/1\right)}{1000}$$

$$R_{\text{lead}} = 0{,}897\ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$$

$$I_{\text{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\text{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\text{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\text{relay}} = 0{,}02\ Ohm$$

Tap ID : 1 (TransformerDifferentialProtection)

Model:RET670, RET650

(Ref: ABB application manual, RET-670, section-2 requirements, Cl.no:2.1.6.1)

Fault Calculations

$$Rated\ primary\ current\ of\ the\ transformer\ (\mathrm{I_{nt}})\ =\ \frac{\mathrm{T_{MVA}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$$

$$\mathrm{I_{nt}}\ =\ \frac{155 * 1000000}{33 * \sqrt{3} * 1000}$$

$$\mathrm{I_{nt}} = 2711{,}7967\ \mathrm{A}$$

$$Through\ fault\ MVA\ of\ the\ Transformer\ \mathrm{T_{MVAe}} = \frac{\mathrm{T_{MVA}} * 100}{\mathrm{Z_t}}$$

$$\mathrm{T_{MVAe}} = \frac{155 * 100}{10}$$

$$\mathrm{T_{MVAe}} = 1550\ MVA$$

$$Maximum\ primary\ fundamental\ frequency\ current\ that\ passes\ two\ main\ CTs\ and\ the\ transformer\ (\mathrm{I_{tf}}) = \frac{\mathrm{T_{MVAe}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = \frac{1550 * 1000000}{33 * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = 27117{,}9672\ A$$

CT knee point voltage Calculation as per Manual

$$Equivalent\ secondary\ e.m.f.for\ close-in\ faults\ (E_{alreq1}) = 30 * I_{nt} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$$

$$E_{alreq1} = 30 * 2711{,}7967 * \frac{1}{3000} * (18 + (2 * 0{,}897) + 0{,}02)$$

$$E_{alreq1}\ = 537{,}3154\ V$$

$$Equivalent\ secondary\ e.m.f.for\ External\ faults\ (E_{alreq2}) = 2 * I_{tf} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} +\ 2R_{lead} + R_{relay}\right)$$

$$E_{alreq2}\ = 2 * 27117{,}9672 * \frac{1}{3000} * (18 + (2 *\ 0{,}897) + 0{,}02)$$

$$E_{alreq2} = 358{,}2103\ V$$

$$Maximum\ secondary\ limiting\ e.m.f\ (V_k) = Max(E_{alreq1}, E_{alreq2})$$

$$V_k = Max(537{,}3154{,}358{,}2103)$$

$$V_k = 537{,}3154\ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$Secondary\ limiting\ e.m.f.\ E_{ALF} = ALF * I_{sr} * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$

$E_{ALF} = 20 * 1 * \left(18 + \frac{20}{1^2}\right)$

$E_{ALF} = 760\ V$

$E_{rq} = E_{ALF}$

$E_{rq} = 760\ V$

$Proposed\ knee\ point\ voltage\ (E_{rq}) > Required\ knee\ point\ voltage (V_k)$

$CT\ Tap\ is\ adequate$

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Summary of Application : TransformerDifferentialProtection

CT-Protection Core

Summary							
Tap Details	***CT Primary current I_{pr} (A)***	***CT Secondary current I_{sr} (A)***	***Accuracy Class***	***Accuracy Limiting Factor (ALF)***	***Knee point voltage(V)***		***Results***
					Proposed	***Calculated***	
Tap-1	3000	1	5PR	20	760	537,3154	Adequate

Core ID: Core-5

Application Name: TransformerDifferentialProtection

Relay Input data

	Primary time constant of the source (three-ph)	T_p	=	110	ms
	Maximum primary fundamental frequency phase to earth current that passed two main CTs without passing the	I_{tf}	=	6000	A
	Transformer rating	T_{MVA}	=	30	MVA
	Transformer impedance at principal tap	Z_t	=	8	%
	Breaker-and-a-half or Double Busbar scheme used	YES/NO	=	No	
	Type of System	$Solidly$ $/High\ impedance$	=	Solidly	

Core data

	Core Details		=	Core-5	
	Core Application		=	TransformerDifferentialProtection	
	Accuracy Class		=	5PR	
	No. of taps on CT Primary	n	=	1	nos
	Relay Make		=	ABB	
	Relay Model		=	RET670	
	Relay Burden	S_R	=	0,02	VA
	Lead Cross sectional area		=	2.5	$Sq.mm$
	Lead D.C resistance at 20 Degree C		=	7,41	Ohm/km
	No. of cable runs	n_1	=	1	nos
	Main cable length		=	40	m
	Cable Conductor Type		=	Copper	

Current transformer Input data

	Description	Symbol		Value	SI Unit
·	**Tap Details**		=	Tap-1	
·	Current Transformer primary current	I_{Pr}	=	500	A
·	Current Transformer nominal secondary current	I_{sr}	=	1	A

·	CT Internal resistance	R_{ct}	=	2	Ohm
·	Rated output of the Current Transformer	S_n	=	10	VA
·	Accuracy limit factor	ALF	=	20	
·	Upper limiting exciting current (mA) (at Vk/2)	I_e	=	0	mA

Lead resistance calculation $(\mathbf{R}_{lead})$**:**

(Ref: IEC 60287-1:2006, Clause no: 2.1, DC resistance of conductor, page no-14)

Lead type: Copper

$$\text{Lead resistance at Max. conductor temperature} = \left(\text{Max. D.C resistance at } 20^0 C * (1 + (Max.conductor\ temp. - 20) * 0.00393)\right)$$

$$\text{Lead resistance at Max. conductor temperature} = (7{,}41 * (1 + (75 - 20) * 0.00393))$$

$$\text{Lead resistance at Max. conductor temperature} = 9{,}0117\ Ohm/kM$$

$$Cable\ Lead\ resistance\ = 2 * \frac{\left((Lead\ resistance\ at\ Max.conductor\ temperature.) * Main\ cable\ length/n_1\right)}{1000})$$

$$R_{\text{lead}} = \ 2 * \frac{\left((9{,}0117) * 40/1\right)}{1000}$$

$$R_{\text{lead}}\ = 0{,}7209\ \ Ohm$$

Relay Parameters Calculation

$$Rated\ Current\ of\ the\ protection\ IED\ \ (I_{\text{R}}) = I_{\text{sr}}$$

$$I_{\text{R}} = 1\ A$$

$$Maximum\ resistance\ of\ the\ protection\ IED\ \ (R_{\text{relay}}) = \frac{S_R}{I_R^2}$$

$$R_{\text{relay}} = \frac{0{,}02}{1^2}$$

$$R_{\text{relay}}\ = 0{,}02\ \ Ohm$$

Tap ID : 1 (TransformerDifferentialProtection)

Model:RET670, RET650

(Ref: ABB application manual, RET-670, section-2 requirements, Cl.no:2.1.6.1)

Fault Calculations

$$Rated\ primary\ current\ of\ the\ transformer\ (\text{I}_{\text{nt}})\ = \frac{\text{T}_{\text{MVA}} * 1000000}{\text{U}_{\text{n}} * \sqrt{3} * 1000}$$

$$\mathrm{I_{nt}} = \frac{30 * 1000000}{33 * \sqrt{3} * 1000}$$

$$\mathrm{I_{nt}} = 524{,}8639\ \mathrm{A}$$

$$Through\ fault\ MVA\ of\ the\ Transformer\ \mathrm{T_{MVAe}} = \frac{\mathrm{T_{MVA}} * 100}{\mathrm{Z_t}}$$

$$\mathrm{T_{MVAe}} = \frac{30 * 100}{8}$$

$$\mathrm{T_{MVAe}} = 375\ MVA$$

$$Maximum\ primary\ fundamental\ frequency\ current\ that\ passes\ two\ main\ CTs\ and\ the\ transformer\ (\mathrm{I_{tf}}) = \frac{\mathrm{T_{MVAe}} * 1000000}{\mathrm{U_n} * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = \frac{375 * 1000000}{33 * \sqrt{3} * 1000}$$

$$\mathrm{I_{tf}} = 6560{,}7985\ A$$

CT knee point voltage Calculation as per Manual

$$Equivalent\ secondary\ e.m.f.for\ close-in\ faults\ (E_{alreq1}) = 30 * I_{nt} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + 2R_{lead} + R_{relay}\right)$$

$$E_{alreq1} = 30 * 524{,}8639 * \frac{1}{500} * (2 + (2 * 0{,}7209) + 0{,}02)$$

$$E_{alreq1} = 109{,}0184\ V$$

$$Equivalent\ secondary\ e.m.f.for\ External\ faults\ (E_{\mathrm{alreq2}}) = 2 * I_{tf} * \frac{I_{sr}}{I_{pr}} * \left(R_{ct} + 2R_{lead} + R_{relay}\right)$$

$$E_{\mathrm{alreq2}} = 2 * 6560{,}7985 * \frac{1}{500} * (2 + (2 * 0{,}7209) + 0{,}02)$$

$$E_{\mathrm{alreq2}} = 90{,}8487\ V$$

$$Maximum\ secondary\ limiting\ e.m.f\ (V_k) = Max(E_{alreq1}, E_{alreq2})$$

$$V_k = Max(109{,}0184, 90{,}8487)$$

$$V_k = 109{,}0184\ V$$

Proposed CT Knee point voltage(conversion of ALF to knee point voltage as per IEC-61869-100)

$$Secondary\ limiting\ e.m.f.\ E_{ALF} = ALF * I_{sr} * \left(R_{ct} + \frac{S_n}{{I_{sr}}^2}\right)$$

$$E_{ALF} = 20 * 1 * \left(2 + \frac{10}{1^2}\right)$$

$$E_{ALF} = 240\ V$$

$$E_{rq} = E_{ALF}$$

$$E_{rq} = 240\ V$$

$Proposed\ knee\ point\ voltage\ (E_{rq}) > Required\ knee\ point\ voltage (V_k)$

$CT\ Tap\ is\ adequate$

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Summary of Application : TransformerDifferentialProtection

CT-Protection Core

Summary							
Tap Details	***CT Primary current*** I_{pr} (A)	***CT Secondary current*** I_{sr} (A)	***Accuracy Class***	***Accuracy Limiting Factor*** (ALF)	***Knee point voltage***(V)		***Results***
					Proposed	***Calculated***	
Tap-1	500	1	5PR	20	240	109,0184	Adequate